Day 11 Study Notes

Mon 23 Nov 2026 · Sprint 60 · JEE Main 2027

Part	Time	What to do
Study	3 h 30 min	Physics: Modern physics (revision) (75 min) · Maths: Sequences and series (part 2) (75 min) · Chemistry: General organic chemistry (part 2) (60 min)
Exam	1 h 15 min	The Day 11 exam in the Sprint 60 app: 30 questions, timed, no notes.
Review	1 h 15 min	Open every wrong and skipped question in the app, read its solution, tag the reason, then use Retry my mistakes.

How to use each chapter note. Read it once without stopping. Cover the formulas and write them from memory. Work the solved examples on paper before reading the working. Then read the NCERT section listed at the end.

Physics · Modern physics (revision)

Modern physics (photoelectric effect, atoms, nuclei, semiconductors) gives four to five questions per paper and is the most scoring part of Physics. Days 4 to 7 covered it chapter by chapter. Today is one sheet with everything together, followed by a mixed test.

1. Photons and the photoelectric effect

- Photon energy: $E = h\nu = \dfrac{hc}{\lambda}$. Shortcut: $E\,(\text{eV}) = \dfrac{1240}{\lambda\,(\text{nm})}$.
- Einstein's equation: $K_{max} = h\nu - \phi = eV_0$, where ϕ is the work function and V_0 the stopping potential.
- Threshold: $\nu_0 = \dfrac{\phi}{h}$, $\lambda_0 = \dfrac{hc}{\phi}$. Below ν_0 no electrons come out, however bright the light.
- **Intensity** changes only the number of electrons (the photocurrent). **Frequency** changes only their maximum energy (the stopping potential).
- Graph of V_0 against ν: a straight line of slope $\dfrac{h}{e}$, the same for every metal, cutting the ν axis at ν_0.

2. Matter waves

- de Broglie: $\lambda = \dfrac{h}{p} = \dfrac{h}{\sqrt{2mK}}$.
- Electron accelerated through V volts: $\lambda = \dfrac{12.27}{\sqrt{V}}$ Å.
- For the same kinetic energy, the lighter particle has the longer wavelength: $\lambda \propto \dfrac{1}{\sqrt{m}}$.

3. Bohr model (hydrogen-like atoms, atomic number Z)

Quantity	Formula	Depends on n as
Radius	$r_n = 0.529\,\dfrac{n^2}{Z}$ Å	n^2
Speed	$v_n = 2.18 \times 10^6\,\dfrac{Z}{n}$ m/s	$\dfrac{1}{n}$
Energy	$E_n = -13.6\,\dfrac{Z^2}{n^2}$ eV	$\dfrac{1}{n^2}$

- Kinetic energy $= -E_n$ and potential energy $= 2E_n$.
- Emitted wavelength: $\dfrac{1}{\lambda} = RZ^2\left(\dfrac{1}{n_1^2} - \dfrac{1}{n_2^2}\right)$, with $\dfrac{1}{R} = 91.2$ nm.
- Series: Lyman ends on $n = 1$ (ultraviolet), Balmer on $n = 2$ (visible), Paschen on $n = 3$ (infrared).
- Number of spectral lines from level n down to the ground state: $\dfrac{n(n-1)}{2}$.

4. Nuclei and radioactivity

- Nuclear radius: $R = R_0 A^{1/3}$, so nuclear density is the same for all nuclei.
- Mass defect Δm gives binding energy $\Delta m \times 931.5$ MeV per u. Binding energy per nucleon peaks near $A = 56$ (iron), which is why heavy nuclei release energy by fission and light ones by fusion.

- Decay law: $N = N_0 e^{-\lambda t}$. Half-life $T_{1/2} = \dfrac{0.693}{\lambda}$. Mean life $\tau = \dfrac{1}{\lambda}$.
- After n half-lives the fraction left is $\left(\dfrac{1}{2}\right)^n$. Activity $A = \lambda N$ falls the same way.
- α decay: A falls by 4, Z by 2. β^- decay: Z rises by 1, A unchanged. γ decay changes neither.

5. Semiconductors and logic gates

- n-type: doped with a group 15 element (pentavalent); electrons are the majority carriers. p-type: a group 13 element (trivalent); holes are the majority carriers. Both are electrically neutral.
- A forward-biased p-n junction conducts; a reverse-biased one carries only a tiny current until breakdown. A Zener diode is used in reverse bias as a voltage regulator.
- Gates: AND $Y = AB$, OR $Y = A + B$, NOT $Y = \bar{A}$, NAND $Y = \overline{AB}$, NOR $Y = \overline{A + B}$. NAND and NOR are universal gates.
- NOR gives 1 only when both inputs are 0. NAND gives 0 only when both inputs are 1.

Solved example 1. Light of wavelength 310 nm falls on a metal of work function 2 eV. Find the stopping potential.
$E = \dfrac{1240}{310} = 4$ eV. $K_{max} = 4 - 2 = 2$ eV, so $V_0 = 2$ V.

Solved example 2. A sample has a half-life of 5 days. What fraction is left after 20 days?
20 days is 4 half-lives, so $\left(\dfrac{1}{2}\right)^4 = \dfrac{1}{16}$ is left.

Traps. Brighter light does not raise the stopping potential. Energy levels of He^+ and Li^{2+} carry the factor Z^2, radii carry $\dfrac{1}{Z}$. The number of lines $\dfrac{n(n-1)}{2}$ is for a group of atoms; a single atom gives at most $n - 1$ lines. Activity and the number of nuclei have the same half-life.

Mathematics · Sequences and series (part 2)

Part 1 covered the AP and GP. Today covers the means, the AM–GM inequality, the AGP and the standard sums. The AM–GM inequality and the method of differences between them give one question in most papers.

1. The three means of two positive numbers a and b

- Arithmetic mean: $A = \dfrac{a+b}{2}$. Geometric mean: $G = \sqrt{ab}$. Harmonic mean: $H = \dfrac{2ab}{a+b}$.
- $A \geq G \geq H$, with equality only when $a = b$.
- $G^2 = AH$, so A, G, H are in GP.
- If A and G are known, a and b are the roots of $x^2 - 2Ax + G^2 = 0$.

2. Inserting means

- n arithmetic means between a and b: $d = \dfrac{b-a}{n+1}$. Their sum is $n \cdot \dfrac{a+b}{2}$.
- n geometric means between a and b: $r = \left(\dfrac{b}{a}\right)^{\frac{1}{n+1}}$. Their product is $\left(\sqrt{ab}\right)^n$.

3. AM–GM inequality

- For positive numbers $x_1, \ldots, x_n$: $\dfrac{x_1 + x_2 + \cdots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \cdots x_n}$.
- **Minimum of a sum with a fixed product:** the sum is least when all terms are equal. Example: for $x > 0$, $x + \dfrac{1}{x} \geq 2$.
- **Maximum of a product with a fixed sum:** the product is greatest when all terms are equal.
- To use it, split the terms so that their product is a constant. Example: $4^x + 4^{1-x} \geq 2\sqrt{4^x \cdot 4^{1-x}} = 4$.

4. Arithmetico-geometric progression (AGP)

- Terms of the form $[a + (k-1)d]\, r^{k-1}$, such as $1 + 2x + 3x^2 + \ldots$
- Method: write S, multiply by r, subtract, and sum the GP that is left.
- Sum to infinity for $|r| < 1$: $S_\infty = \dfrac{a}{1-r} + \dfrac{dr}{(1-r)^2}$. In particular $1 + 2x + 3x^2 + \cdots = \dfrac{1}{(1-x)^2}$.

5. Standard sums

Sum	Value	$n = 10$
$\sum k$	$\dfrac{n(n+1)}{2}$	55
$\sum k^2$	$\dfrac{n(n+1)(2n+1)}{6}$	385
$\sum k^3$	$\left[\dfrac{n(n+1)}{2}\right]^2$	3025

- For a series whose kth term is a polynomial in k, write T_k and sum each power separately. Example: $\sum k(k+1) = \sum k^2 + \sum k = \dfrac{n(n+1)(n+2)}{3}$.

- **Method of differences (telescoping):** write $T_k = f(k) - f(k+1)$ and most terms cancel. Example: $\dfrac{1}{k(k+1)} = \dfrac{1}{k} - \dfrac{1}{k+1}$, so $\sum_{k=1}^{n} \dfrac{1}{k(k+1)} = 1 - \dfrac{1}{n+1} = \dfrac{n}{n+1}$.
- If the differences of the terms form an AP, the nth term is a quadratic in n; if they form a GP, it has the form $a\,r^n + b$.

Solved example 1. The AM of two positive numbers is 5 and their GM is 4. Find the numbers and their HM.

They are the roots of $x^2 - 10x + 16 = 0$: 8 and 2. $H = \dfrac{G^2}{A} = \dfrac{16}{5} = 3.2$.

Solved example 2. Find $\dfrac{1}{1\cdot 3} + \dfrac{1}{3\cdot 5} + \ldots$ to n terms.

$\dfrac{1}{(2k-1)(2k+1)} = \dfrac{1}{2}\left(\dfrac{1}{2k-1} - \dfrac{1}{2k+1}\right)$, so the sum is $\dfrac{1}{2}\left(1 - \dfrac{1}{2n+1}\right) = \dfrac{n}{2n+1}$.

Traps. AM–GM needs positive terms. Equality, and so the minimum, needs the terms to be able to be equal. $\sum k^3$ is the square of $\sum k$, not of $\sum k^2$. In telescoping sums, check the first and last terms that survive.

Chemistry · General organic chemistry (part 2)

Part 1 covered the electronic effects. Today uses them to answer the three questions JEE asks most often from this chapter: which acid is stronger, which base is stronger, and which intermediate is more stable. Aromaticity is also here.

1. Acidity

- An acid is stronger when its conjugate base (the anion left after losing H^+) is more stable.
- Anything that spreads out the negative charge makes the acid stronger: −I groups, −R groups, resonance, more s character.
- General order: sulphonic acids > carboxylic acids > phenol > water > alcohols > terminal alkynes.
- **Phenols:** $-NO_2$, −CN, −Cl make phenol more acidic; $-CH_3$ and $-OCH_3$ make it less acidic. A nitro group at ortho or para (where it can withdraw by resonance) has a much bigger effect than at meta. p-Nitrophenol > o-nitrophenol > m-nitrophenol > phenol > p-cresol.
- **Carboxylic acids:** $Cl_3CCOOH > Cl_2CHCOOH > ClCH_2COOH > HCOOH > CH_3COOH$. Benzoic acid ($pK_a$ 4.2) is stronger than acetic acid (4.76).
- **Ortho effect:** almost any group at the ortho position makes benzoic acid stronger.
- **Alcohols:** methanol > primary > secondary > tertiary, because alkyl groups push electrons onto the alkoxide oxygen.

2. Basicity

- A base is stronger when its lone pair is more available to take up a proton.
- **Aliphatic amines in water:** $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N > NH_3$. The +I effect raises basicity, but solvation of the protonated amine and crowding lower it for the tertiary amine. With ethyl groups: $(C_2H_5)_2NH > (C_2H_5)_3N > C_2H_5NH_2 > NH_3$.
- **In the gas phase** there is no solvation, so the order follows the +I effect: tertiary > secondary > primary > NH_3.
- **Aniline** is a much weaker base than ammonia: its lone pair is delocalised into the ring. $-NO_2$ on the ring weakens it further; $-CH_3$ and $-OCH_3$ at para strengthen it.
- Lone pair on an sp^3 nitrogen is more basic than on sp^2, which is more basic than on sp: amines > pyridine > nitriles.
- If the lone pair is part of an aromatic sextet (pyrrole), the compound is hardly basic at all.

3. Reaction intermediates

Intermediate	Carbon	Shape	Stabilised by	Order of stability
Carbocation	sp^2, 6 electrons	planar	+I, hyperconjugation, +R	benzyl, allyl ≈ 3° > 2° > 1° > CH_3^+
Free radical	sp^2, 7 electrons	planar	same as carbocations	benzyl, allyl > 3° > 2° > 1° > CH_3•
Carbanion	sp^3, 8 electrons	pyramidal	−I, −R	CH_3^- > 1° > 2° > 3°
Carbene	two bonds, 6 electrons	bent or linear	—	neutral, very reactive

Carbanions follow the reverse order of carbocations, because alkyl groups push more electrons onto a carbon that is already negative.

4. Aromaticity (Hückel's rule)

- A species is **aromatic** if it is cyclic, planar, fully conjugated (every ring atom has a p orbital) and has $(4n + 2)$ π electrons: 2, 6, 10, 14.
- With $4n$ π electrons (4, 8) and all other conditions met, it is **antiaromatic** and very unstable.
- Aromatic: benzene, naphthalene (10 π), pyridine, pyrrole, furan, thiophene, cyclopropenyl cation (2 π), cyclopentadienyl anion (6 π), cycloheptatrienyl (tropylium) cation (6 π).
- Not aromatic: cyclopentadiene (one sp^3 carbon breaks the conjugation), cyclooctatetraene (tub-shaped, not planar).
- Antiaromatic: cyclobutadiene, cyclopentadienyl cation (4 π).

5. Types of organic reactions

- **Substitution:** one group replaces another ($CH_3Br + OH^- \rightarrow CH_3OH$).
- **Addition:** a multiple bond takes up atoms (ethene + HBr).
- **Elimination:** atoms are removed to form a multiple bond (alkyl halide + alcoholic KOH → alkene).
- **Rearrangement:** the carbon skeleton or a group shifts, often through a carbocation moving to a more stable one.

> **Solved example 1.** Arrange in increasing acidity: phenol, ethanol, acetic acid.
> Ethanol < phenol < acetic acid. The phenoxide ion is stabilised by resonance into the ring; the acetate ion spreads its charge over two equally electronegative oxygens.

> **Solved example 2.** Is the cyclopentadienyl anion aromatic?
> It is cyclic and planar, every carbon is sp^2 once the lone pair joins the ring, and it has 6 π electrons. So yes.

Traps. Aqueous and gas-phase amine orders differ; read which one is asked. Carbanion stability runs opposite to carbocation stability. Cyclopentadiene itself is not aromatic; its anion is. Meta-nitrophenol gains only the −I effect, not −R.